

CS5707 Artificial Intelligence
Introduction

Yongmin Li

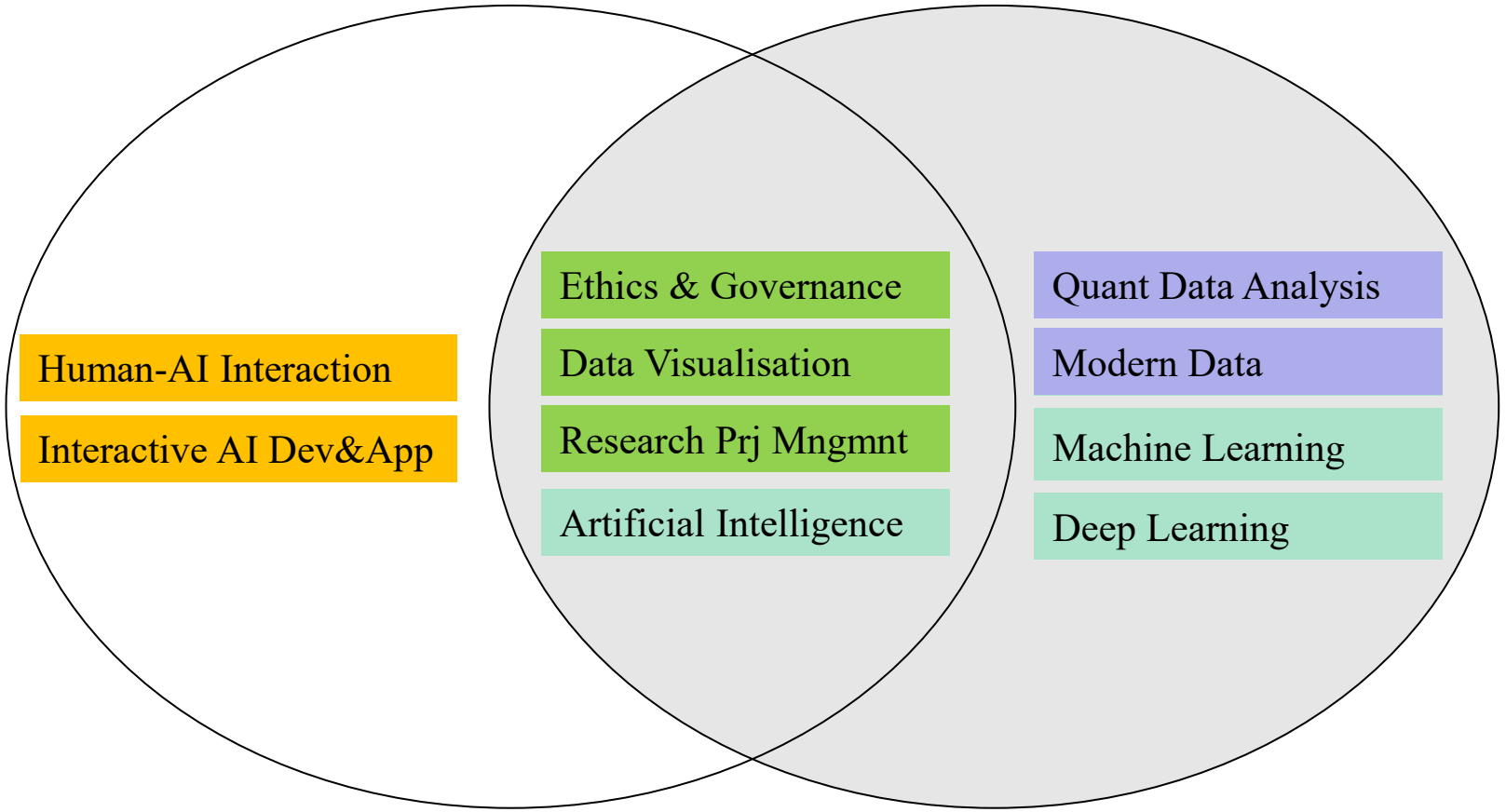
Lecturer

Prof. Yongmin Li
http://www.brunel.ac.uk/~csstyyl/
Email: Yongmin.Li@brunel.ac.uk

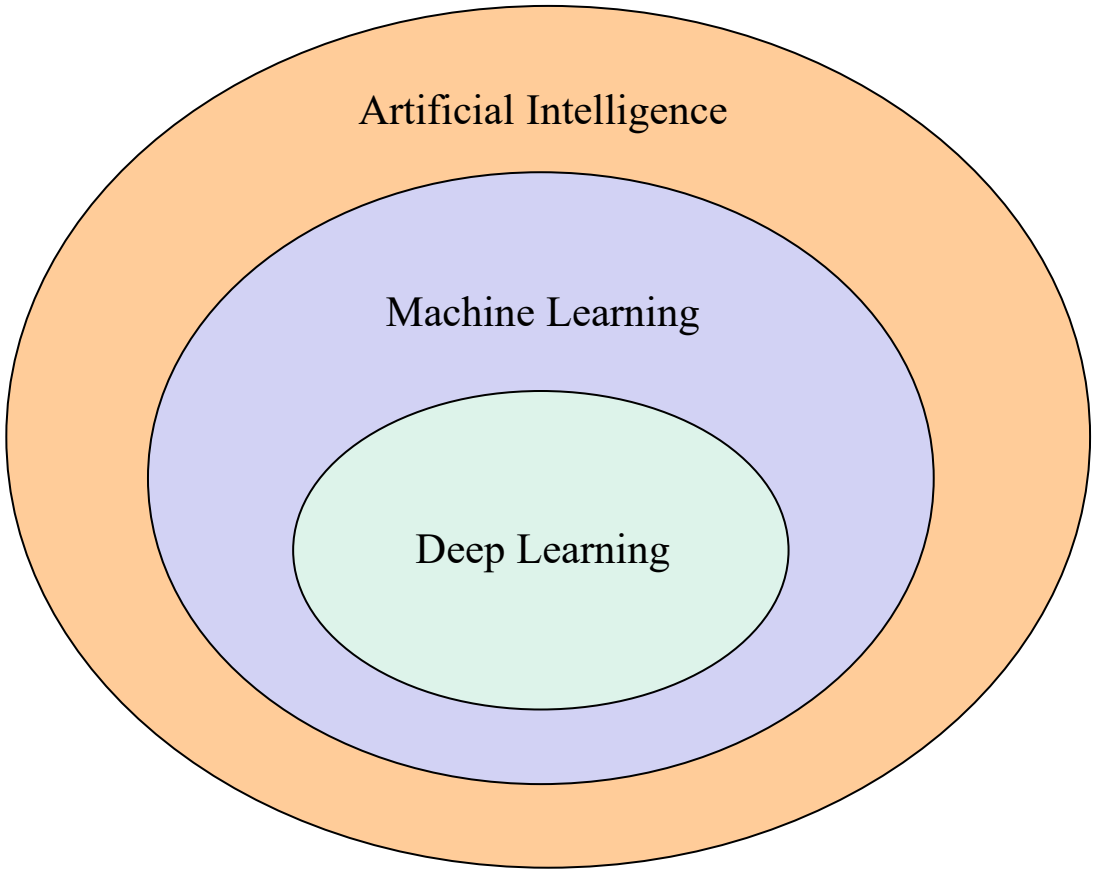
Overview

- **MSc AI/HAII programmes**
- **Module**
- **Method of assessment**
- **Artificial Intelligence (AI)**
- **Applications of AI**

HAII vs AI

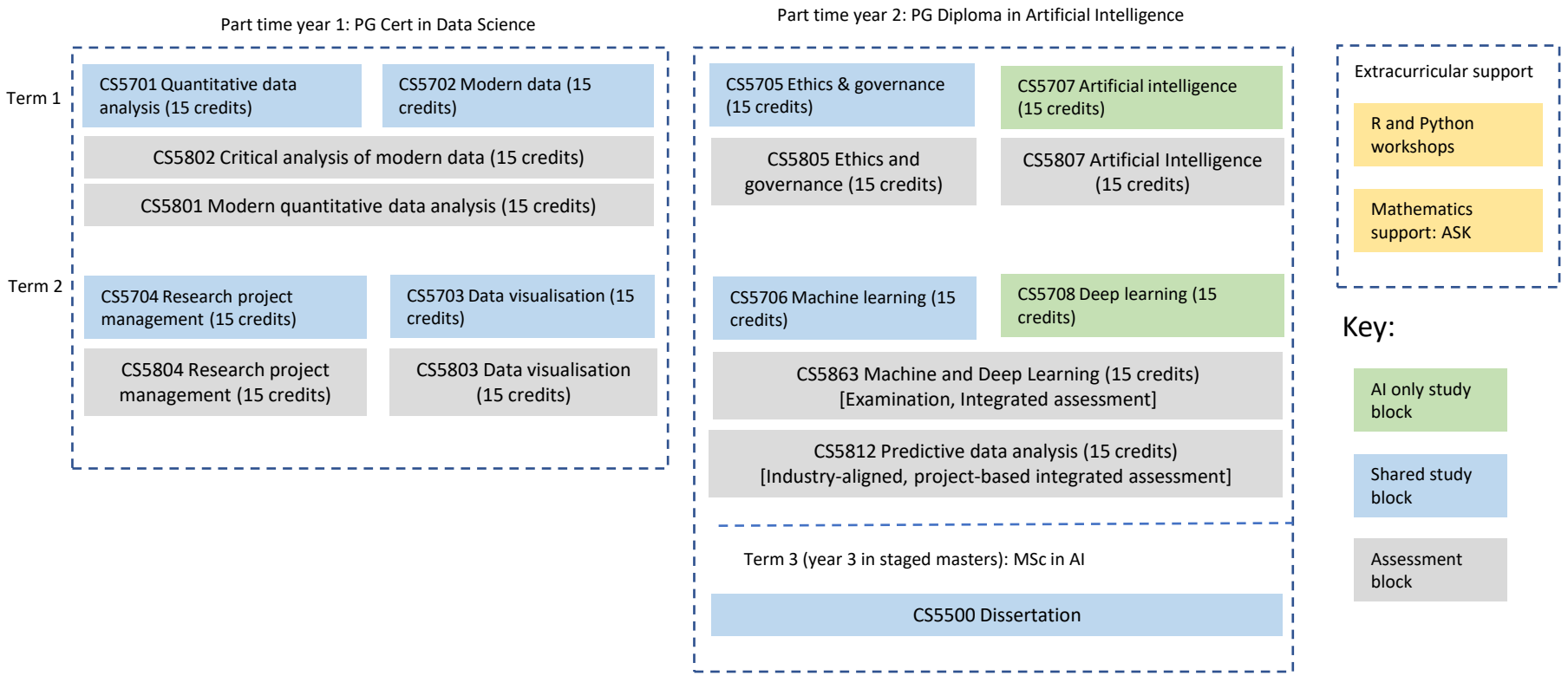

Human-AI Interaction
Interactive AI Dev&App
Ethics & Governance
Data Visualisation
Research Prj Mngmnt
Artificial Intelligence
Quant Data Analysis
Modern Data
Machine Learning
Deep Learning
MSc Human AI Interaction
MSc Artificial Intelligence

AI vs ML vs DL



Artificial Intelligence, Machine Learning and Deep Learning

Study Blocks & Assessment Blocks



Topics

1. Introduction
2. Search
3. Genetic Algorithms
4. Probabilistic Inference
5. Fuzzy Systems
6. Image and Vision
7. Natural Language Processing
8. Large Language Models
9. Generative AI

Learning Outcomes

- ✓ **Develop an in-depth knowledge and understanding of the key concepts of artificial intelligence and fundamental methods of artificial intelligence.**

- ✓ **Learn how to apply appropriate artificial intelligence methods to defined problems, justify the approach adopted, and critically evaluate its effectiveness**

My Personal Goal

- **Most interesting**
- **Most challenging**
- **Most learned**

Sessions

Lectures

- **Thursdays 12:00-13:00**
- **LECT 266**

Labs/Seminars

- **Thursdays 15:00-17:00**
- **TOWC 402**

Assessment

Examination

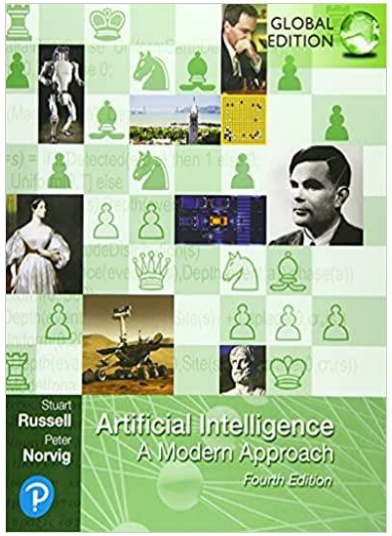
- 100% of the overall marks
- 2 hours
- Winter exam period (exact date TBC)

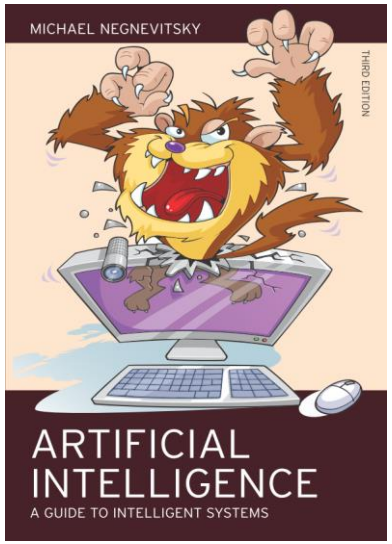
Reading

[1] Russell, S. and Norvig, P. (2021). Artificial Intelligence: A Modern Approach. 4th Edition. Pearson Education Limited. (3rd edition 2014 is fine as well)

[2] Negnevitsky M. (2011). Artificial Intelligence: A Guide to Intelligent Systems. 3rd Edition. Addison Wesley.

Modern AI: recent papers.





Be Prepared…

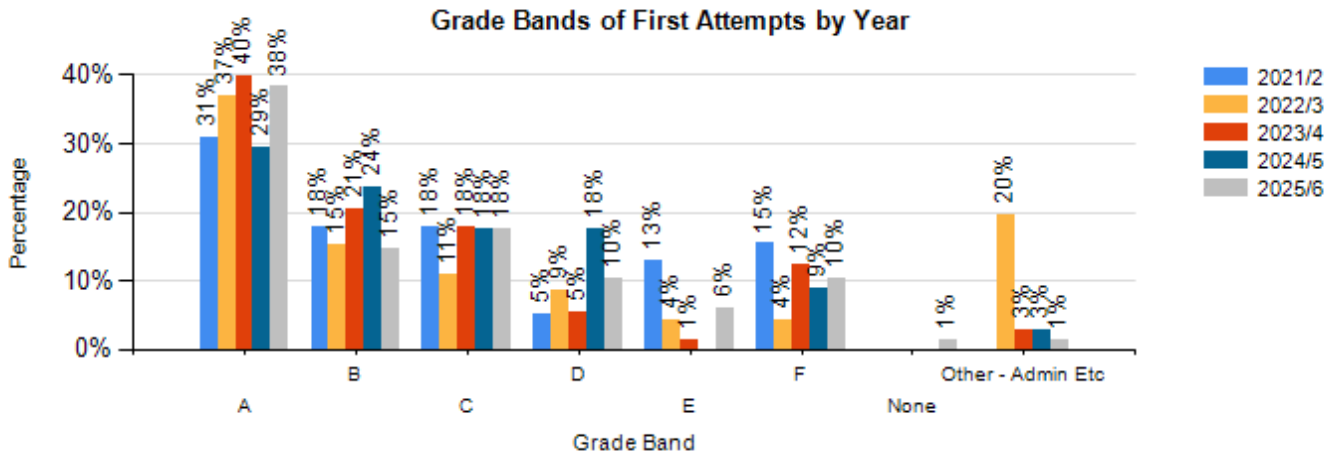
This may be the hardest module in the course!

- International benchmarks
- Market demand & standards
- Coverage of topics
- Highly applicable methods → practical skills

However…

- You will do fine
- I will help you

CS5707 Assessment Results



First attempt results over the past 5 years

What Is Intelligence

- **‘intelligence: the faculty of understanding; intellect [the faculty of knowing and reasoning]; quickness or superiority of understanding, sagacity [acuteness of mental discernment; soundness of judgement, shrewdness]; the action or fact of understanding something; knowledge, comprehension (of something). *(Oxford Dictionary).***
- **‘intelligence: the ability to comprehend; to understand and profit from experience; intelligent: having the capacity for thought and reason especially to a high degree (‘is there intelligent life in the universe?’). *(WordNet 1.7).***
- **‘intelligence: the capacity to acquire and apply knowledge; the faculty of thought and reason. *(The American Heritage® Dictionary).***
- **‘intelligence: the ability to learn, understand and make judgments or have opinions that are based on reason. *(Cambridge International Dictionary).***
- **‘intelligence: the ability to learn or understand or to deal with new or trying situations; reason, also the skilled use of reason; the ability to apply knowledge to manipulate one’s environment or to think abstractly as measured by objective criteria (as tests); mental acuteness, shrewdness; the act of understanding, comprehension. *(Merriam-Webster Dictionary).***

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Artificial Intelligence

Artificial Intelligence (AI) is a science that has defined its goal as making machines do things that would require intelligence if done by humans.

Negnevitsky 2004

The scientific understanding of the mechanisms underlying thought and intelligent behaviour and their embodiment in machines.

American Association for Artificial Intelligence

In this module…

In this module, we will be discussing

- ✓ search
- ✓ genetic algorithms
- ✓ probabilistic inference
- ✓ fuzzy systems
- ✓ image processing
- ✓ natural language processing
- ✓ interactive AI
- ✓ generative AI

We will NOT talk about (but will do in other modules)

- ethics of AI
- machine learning
- deep learning

Applications of AI

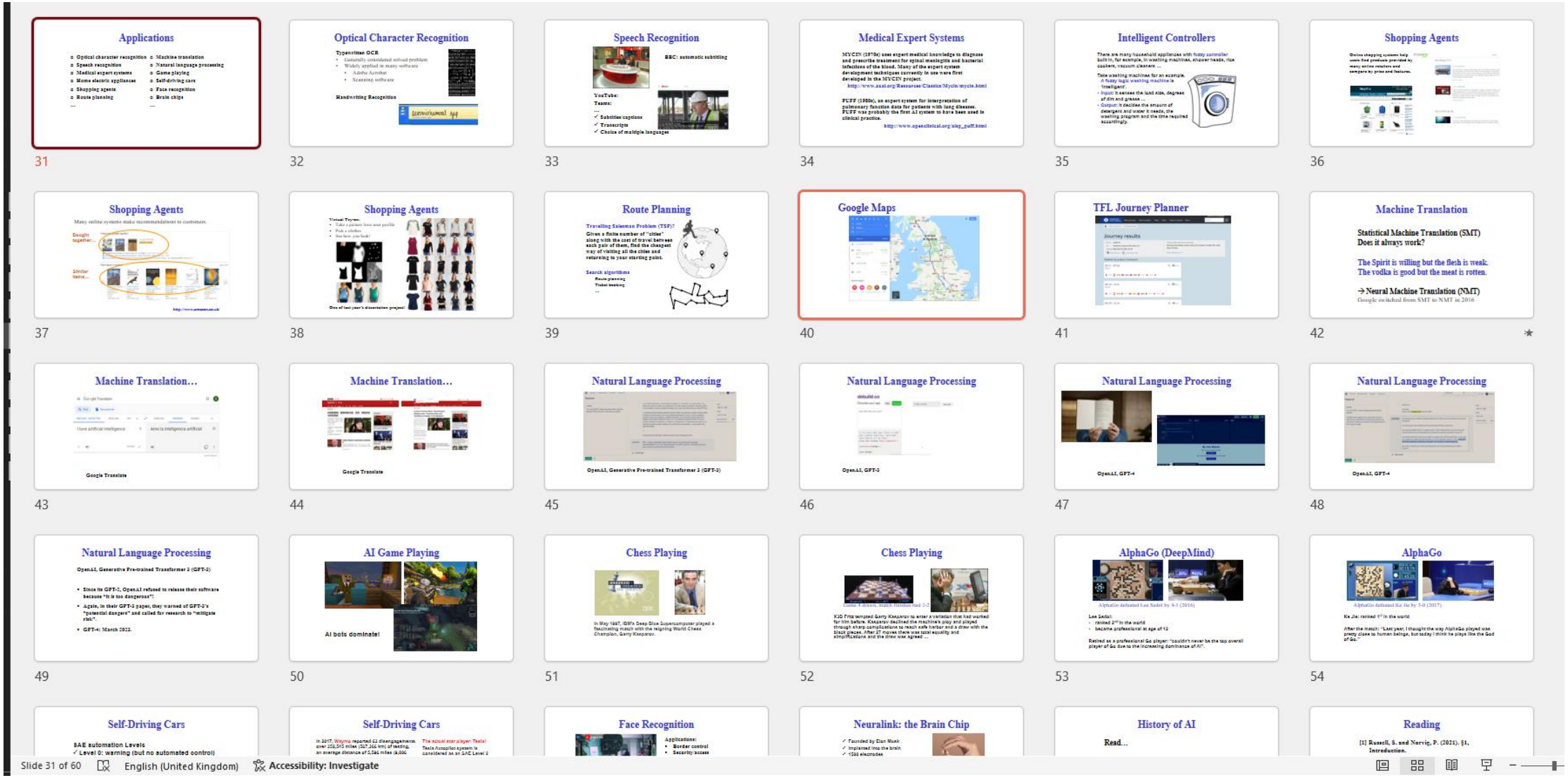
- Optical character recognition
- Speech recognition
- Medical expert systems
- Home electric appliances
- Shopping agents
- Route planning

…

- Machine translation
- Natural language processing
- Game playing
- Self-driving cars
- Face recognition
- Brain chips

…

Early Years…


Applications
Optical Character Recognition
Speech Recognition
Medical Expert Systems
Intelligent Controllers
Shopping Agents
31
32
33
34
35
36
Shopping Agents
Shopping Agents
Route Planning
Google Maps
TFL Journey Planner
Machine Translation
Statistical Machine Translation (SMT)
Does it always work?
The Spirit is willing but the flesh is weak.
The vodka is good but the meat is rotten.
→Neural Machine Translation (NMT)
37
38
39
40
41
42
Machine Translation…
Machine Translation…
Natural Language Processing
Natural Language Processing
Natural Language Processing
Natural Language Processing
43
44
45
46
47
48
Natural Language Processing
AI Game Playing
Chess Playing
Chess Playing
AlphaGo (DeepMind)
AlphaGo
49
50
51
52
53
54
Self-Driving Cars
Self-Driving Cars
Face Recognition
Neuralink: the Brain Chip
History of AI
Reading

Applications of AI

- Tasks within the module
- Dissertation projects

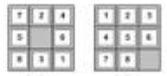
Task Within the Module

1 — Route Planning



Romania: Travel from Arad to Bucharest

2 — Classic Puzzles



The 8-puzzle problem

3 — Solving a Maze



Maze solving problem

4 — Travelling Salesman Problem



A salesman has to visit all N cities 1,2,...,N. Assuming the distances between cities are known, find the best sequence of cities to minimise the travel distance.

5 — The 8-Queens Problem



The 8-queens problem

6 — The Map Colouring Problem



The map colouring problem

7 — The Iris Classification Problem



The Iris classification problem

8 — Disease Diagnosis



Predict the progression level of diabetes

9 — The Monty Hall Problem



The Monty Hall problem

10 — The Tipping Problem



Determine tips by quality of food and service

11 — Project Risk Assessment



Evaluate the risk of a project from funding, staffing etc.

12 — Washing Machine Controller



Determine washing programs from dirtiness and load of clothes

13 — Sentiment Analysis



Analyse the polarity of movie reviews, tweets etc

14 — Text Classification



Classify documents/messages into pre-determined categories

15 — Topic Modelling



Discover hidden semantic structure from large corpus of text

16 — Face Detection



Detect faces using the Haar cascade face detection method

17 — Image Segmentation



Instance segmentation using R-CNN

18 — Object Recognition

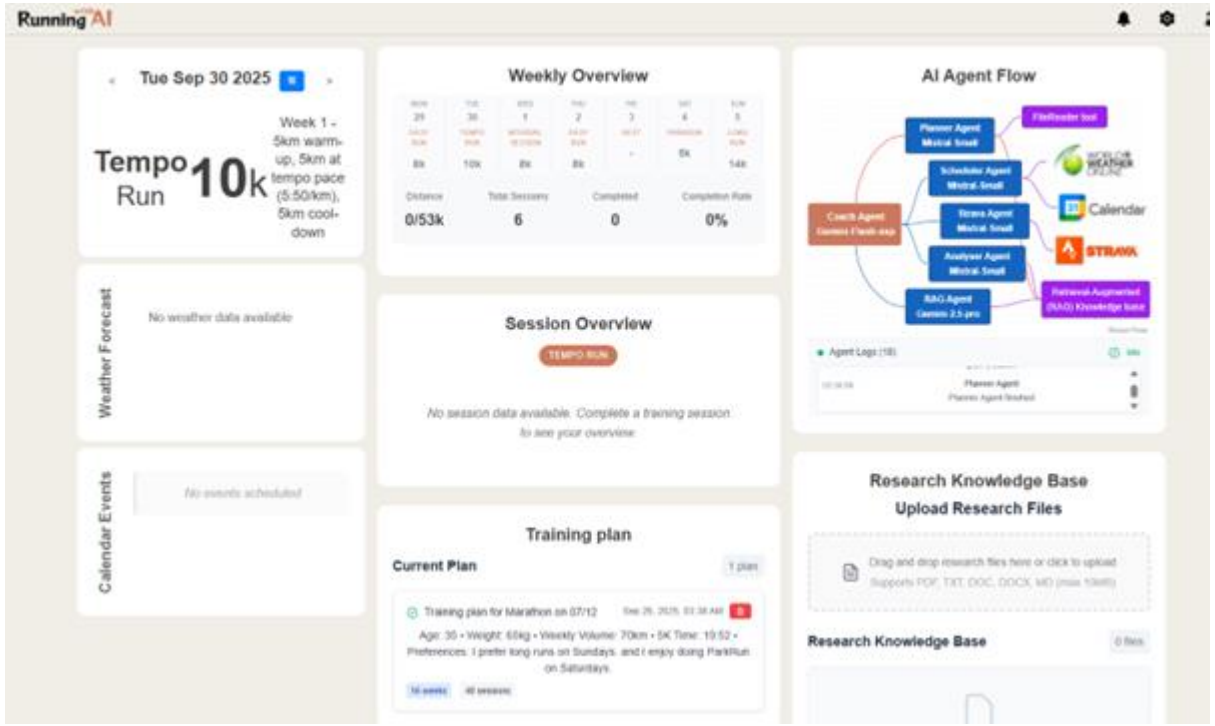


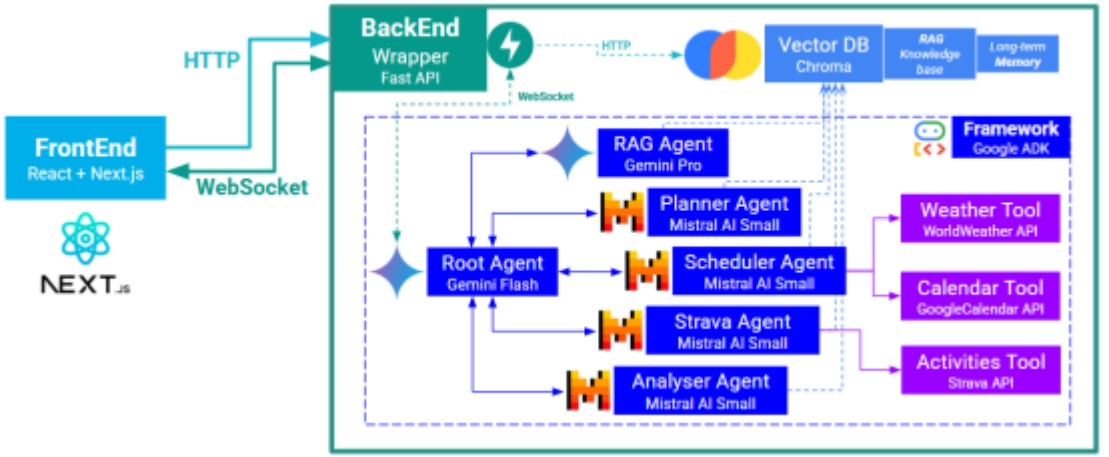
Object recognition from images using YOLO

Three tasks per week – model solutions provided!

Past Dissertations

Dissertation: RunningAI

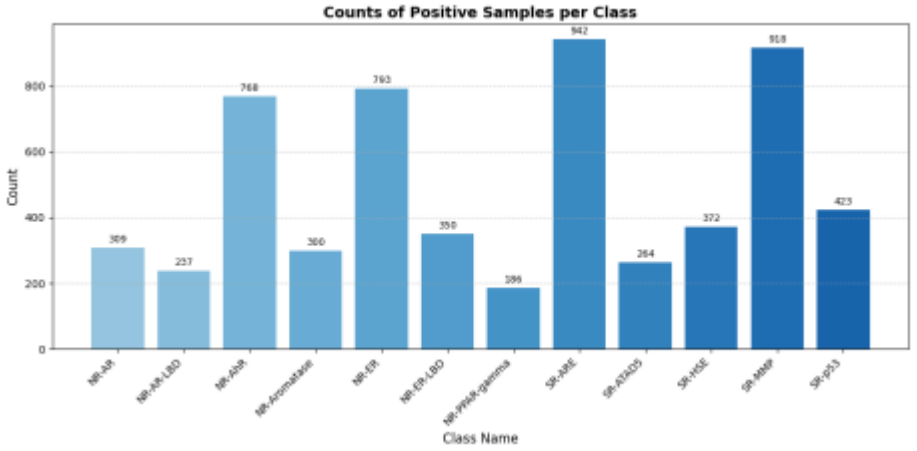


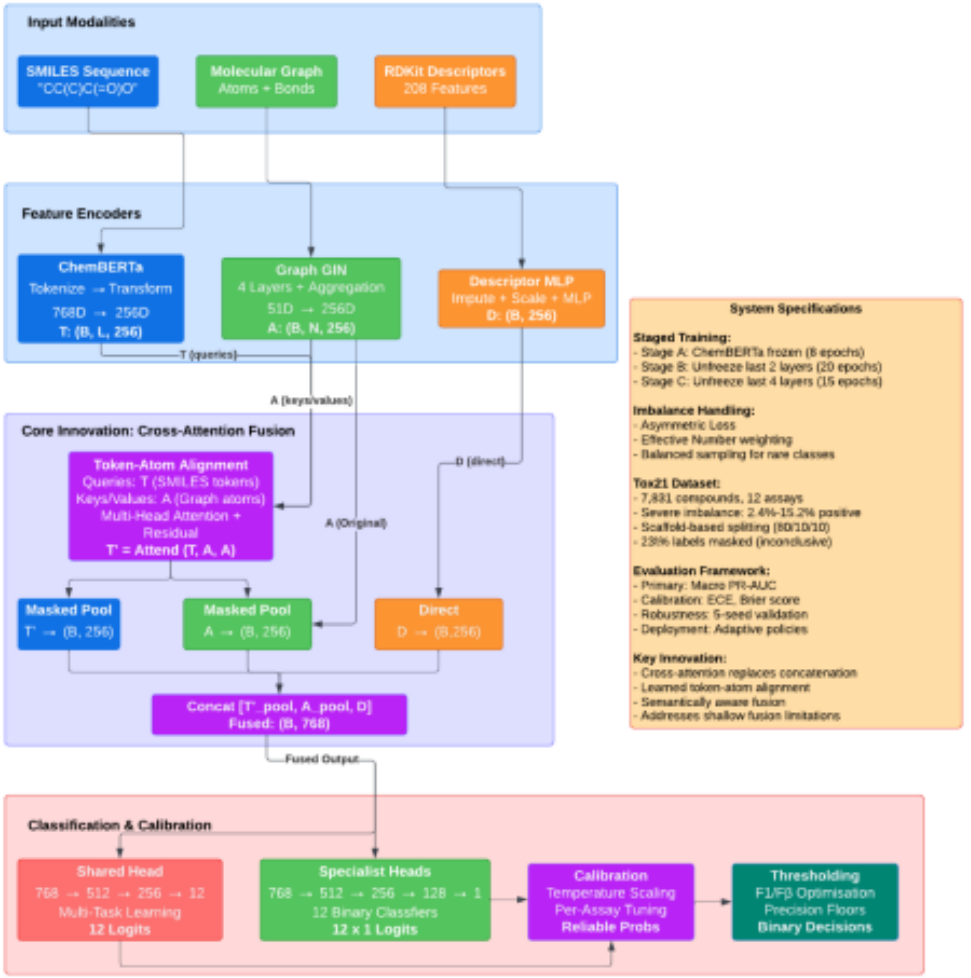


- Personalised coaching for runners
- Agentic AI
- Contextual awareness: calendar, weather, running activities
- Retrieval-Augmented Generation (RAG)
- Training plans
- Session analysis & feedback

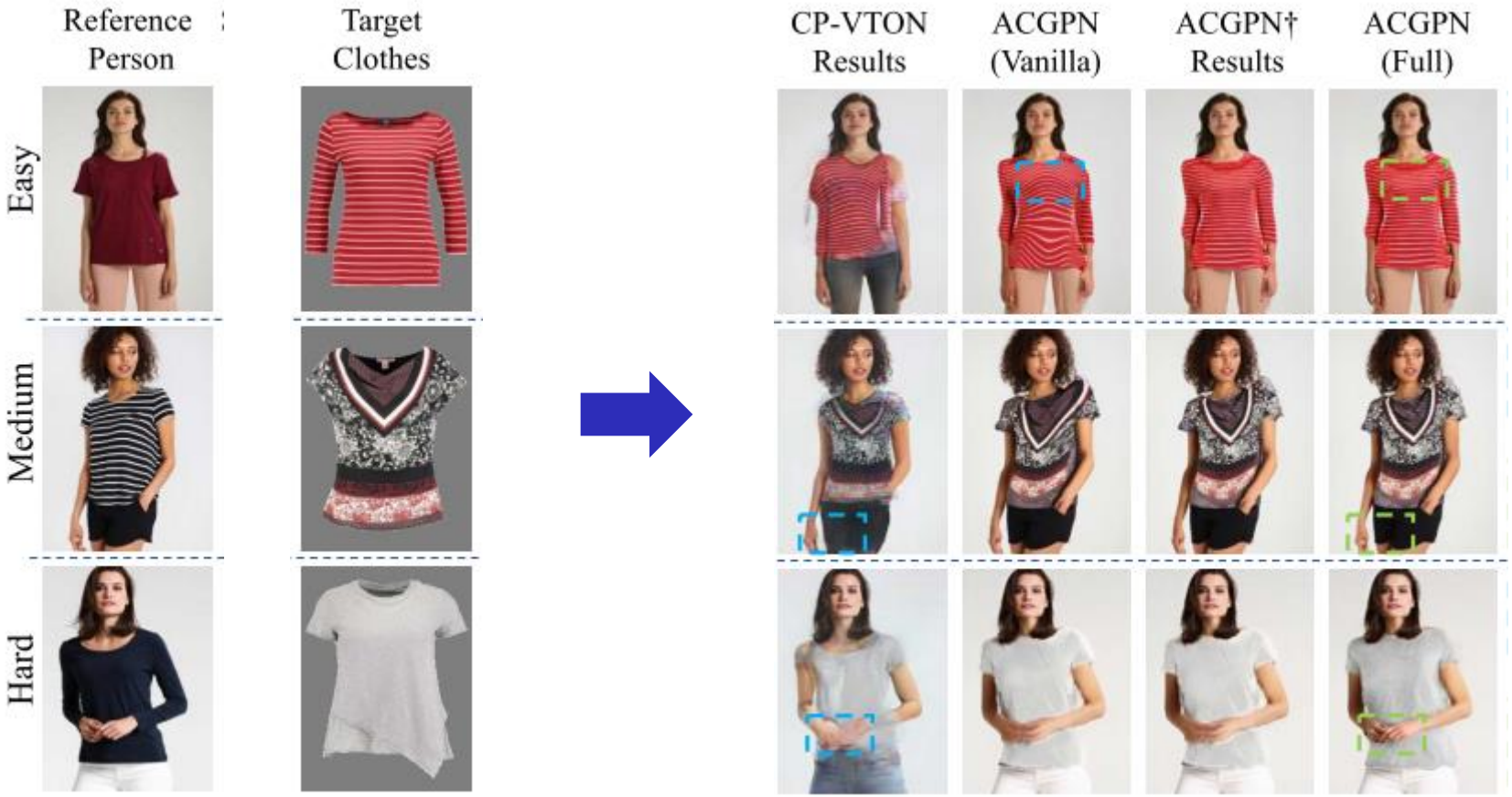
Dissertation: Molecular Toxicity Prediction

- Multi-modal transformer:
  - SMILES sequences
  - Molecular graphs
  - Physiochemical descriptors
- Tox21 dataset: 7831 compounds with 12 toxicity assays
- Class imbalance

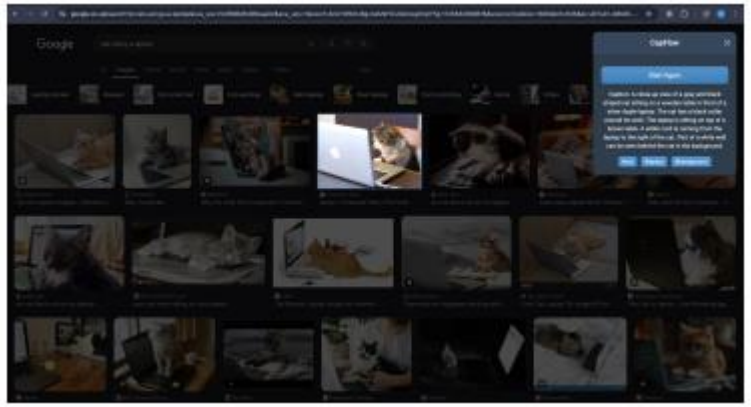




Dissertation: Virtual Try-On

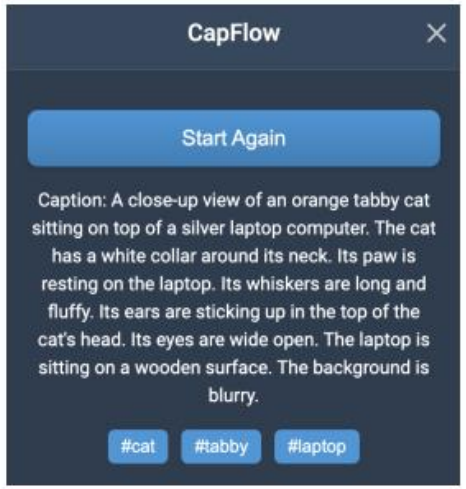


Dissertation: Image Captioning





(a) Original Image



(b) CapFlow Extension Output

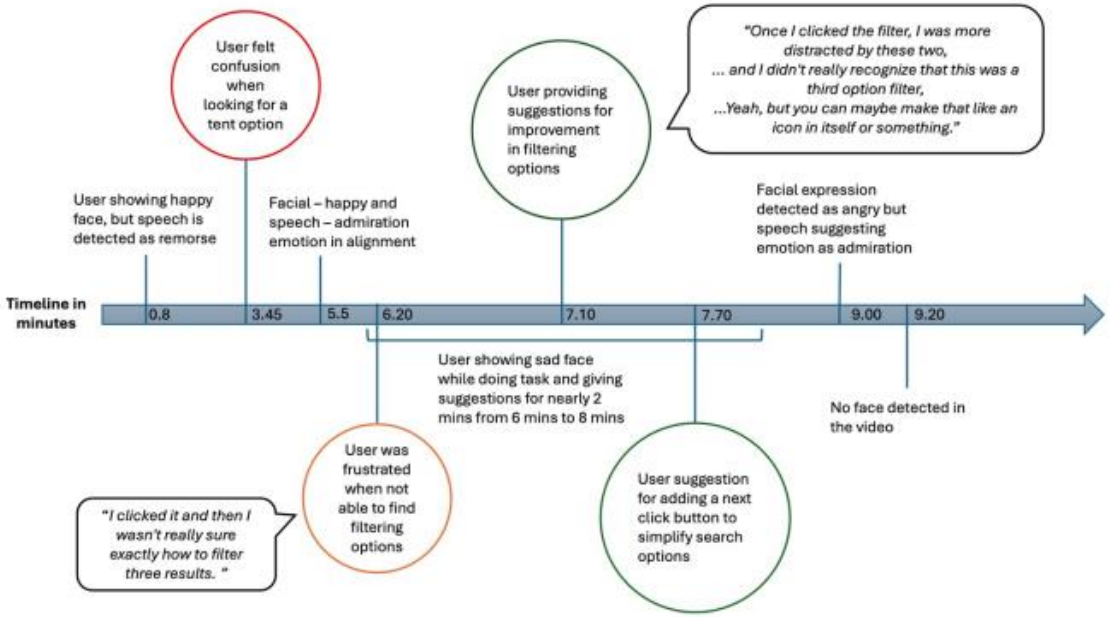
A Chrome extension:

- Automatically generate the descriptions of a selected image (image captioning)
- Further provide a few hashtags.

Shetty, N. and Li, Y. (2024) 'Detailed image captioning and hashtag generation', Future Internet, 16 (12), p. 444.

Dissertation: Insight from Video Reviews

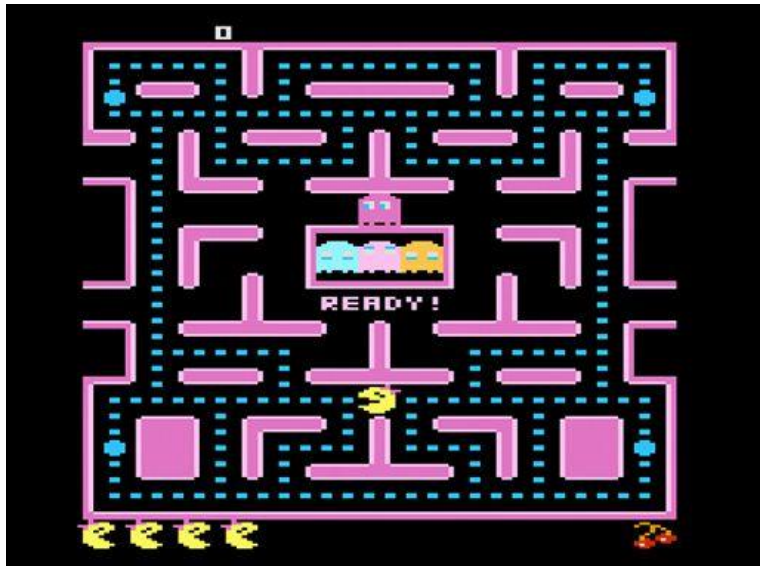


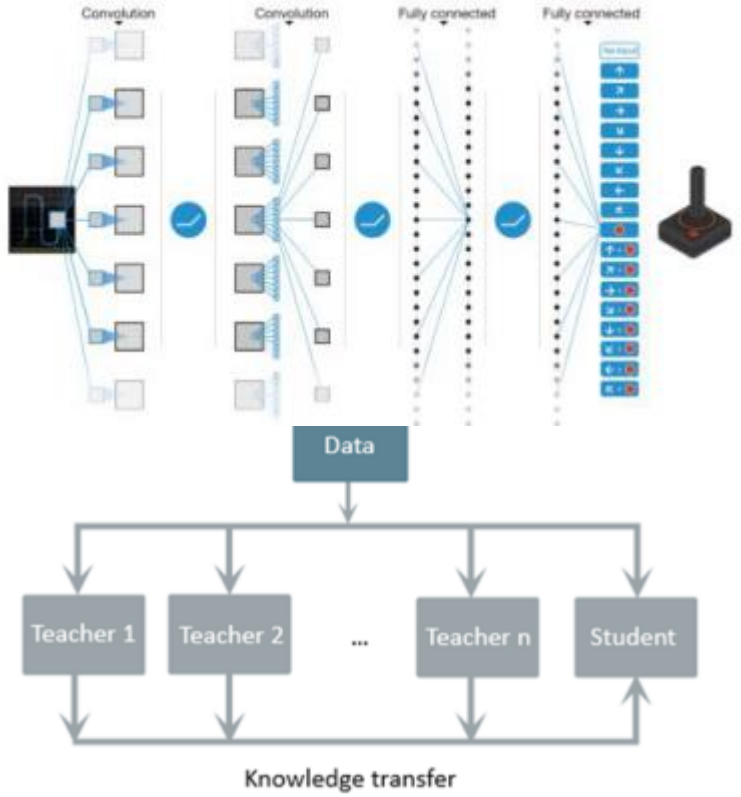


- Video: expression recognition of the reviewer
- Audio: sentiment analysis for the subtitles (text)
- Insight: timeline visualisation of key issues in the video review.

Ghatoray, S.K. and Li, Y. (2026) 'Extracting meaningful insights from user research videos', International Journal of Human–Computer Interaction, pp. 1–19.

Dissertation: Gaming





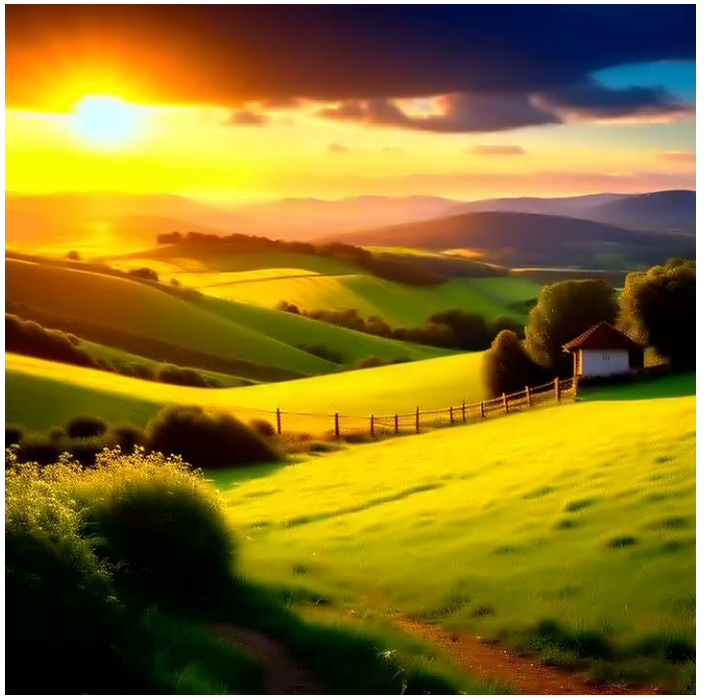
- Reinforcement Learning
- Deep Q-Learning
- Knowledge Distillation: from teachers to student model

Dissertation: Music Video





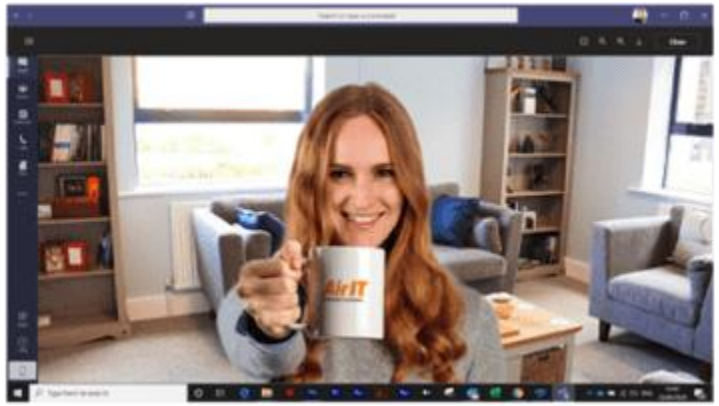




Generate video from: music + initial image of your choice

- Music to text (captioning)
- Image Generation (diffusion models)
- Audio controllers (harmonics and percussive)
- Video synthesis

Dissertation: Virtual Background

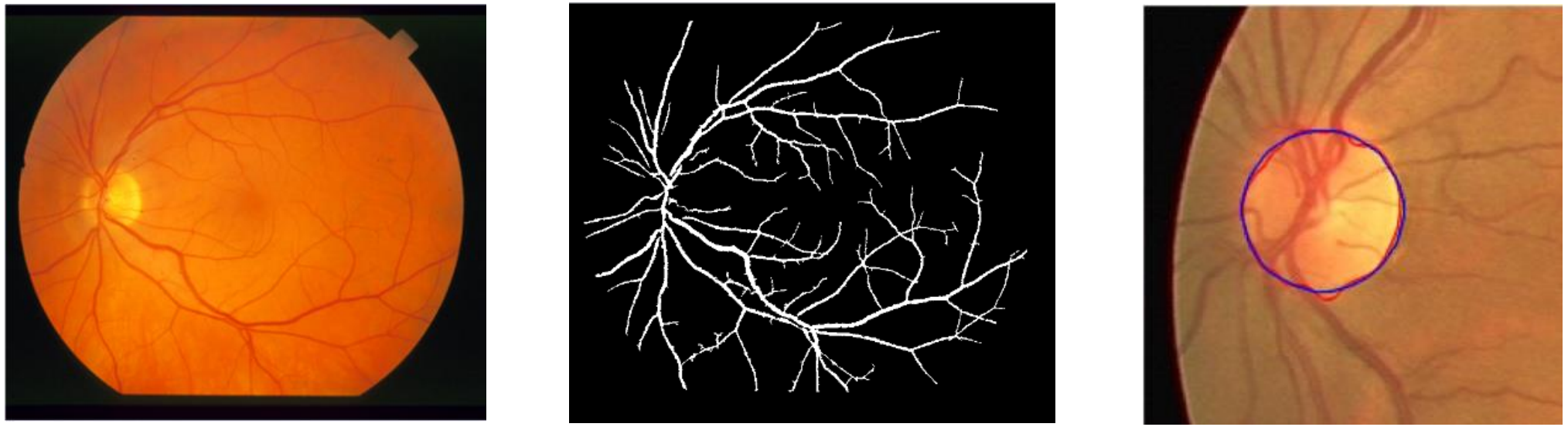




Used for video conferencing or online meeting: Teams, Zoom, Meet…
Essentially a segmentation problem: separate the person from the background

Shahi, K. and Li, Y. (2023) 'Background replacement in video conferencing', International Journal of Network Dynamics and Intelligence, 2 (2), pp. 1–11.

Dissertation: Medical Imaging



2D Fundus Images: Blood Vessels & Optic Disc

Ehab, W., Huang, L. and Li, Y. (2024) 'UNet and variants for medical image segmentation', International Journal of Network Dynamics and Intelligence, 3 (2), 100009, pp. 1–23.

Dissertation: NLP for Tax Assessment

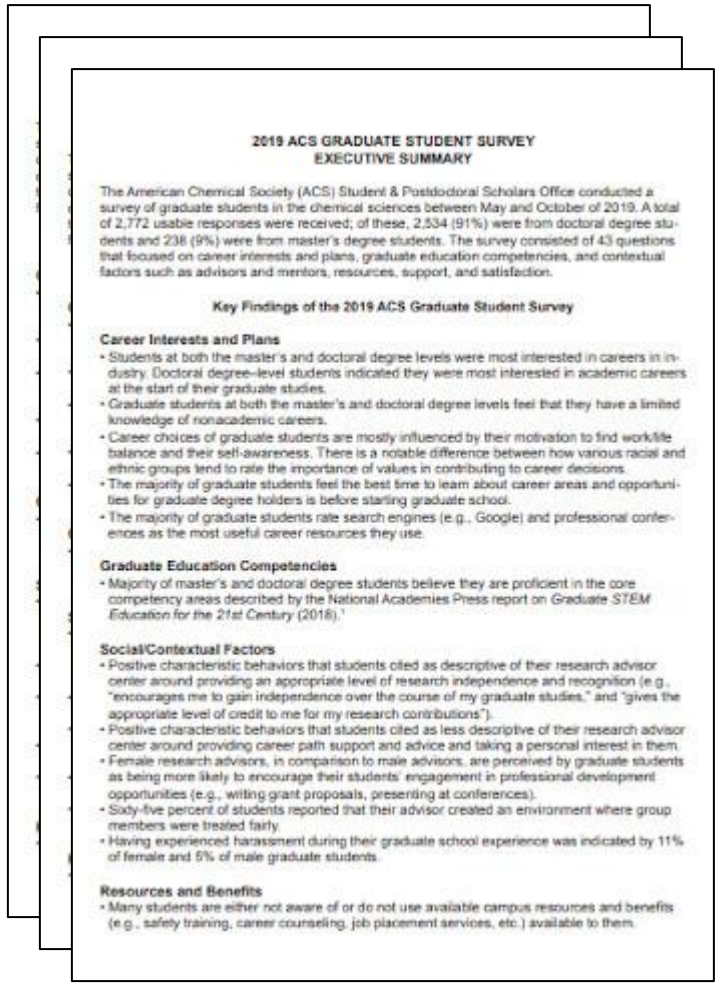
Date	Description	Ref.	Withdrawals	Deposits	Balance
2003-10-08	Previous balance				0.55
2003-10-14	Payroll Deposit - HOTEL			694.81	695.36
2003-10-14	Web Bill Payment - MASTERCARD	9685	200.00		495.36
2003-10-16	ATM Withdrawal - INTERAC	3990	21.25		474.11
2003-10-16	Fees - Interac		1.50		472.61
2003-10-20	Interac Purchase - ELECTRONICS	1975	2.99		469.62
2003-10-21	Web Bill Payment - AMEX	3314	300.00		169.62
2003-10-22	ATM Withdrawal - FIRST BANK	0064	100.00		69.62
2003-10-23	Interac Purchase - SUPERMARKET	1559	29.08		40.54
2003-10-24	Interac Refund - ELECTRONICS	1975		2.99	43.53
2003-10-27	Telephone Bill Payment - VISA	2475	6.77		36.76
2003-10-28	Payroll Deposit - HOTEL			694.81	731.57
2003-10-30	Web Funds Transfer - From SAVINGS	2620		50.00	781.57
2003-11-03	Pre-Auth. Payment - INSURANCE		33.55		748.02
2003-11-03	Cheque No. - 409		100.00		648.02
2003-11-06	Mortgage Payment		710.49		-62.47
2003-11-07	Fees - Overdraft		5.00		-67.47
2003-11-08	Fees - Monthly		5.00		-72.47
	*** Totals ***		1,515.63	1,442.61	

Transaction descriptions
➔Tax Codes:
Automatically tagging.

Ledgers/contracts items: data generated for accounting, but not for tax purpose!

Dissertation: Automated NSS Analysis

National Student Survey



2019 ACS GRADUATE STUDENT SURVEY
EXECUTIVE SUMMARY

The American Chemical Society (ACS) Student & Postdoctoral Scholars Office conducted a survey of graduate students in the chemical sciences between May and October of 2019. A total of 2,772 usable responses were received; of these, 2,534 (91%) were from doctoral degree students and 238 (9%) were from master's degree students. The survey consisted of 43 questions that focused on career interests and plans, graduate education competencies, and contextual factors such as advisors and mentors, resources, support, and satisfaction.

Key Findings of the 2019 ACS Graduate Student Survey

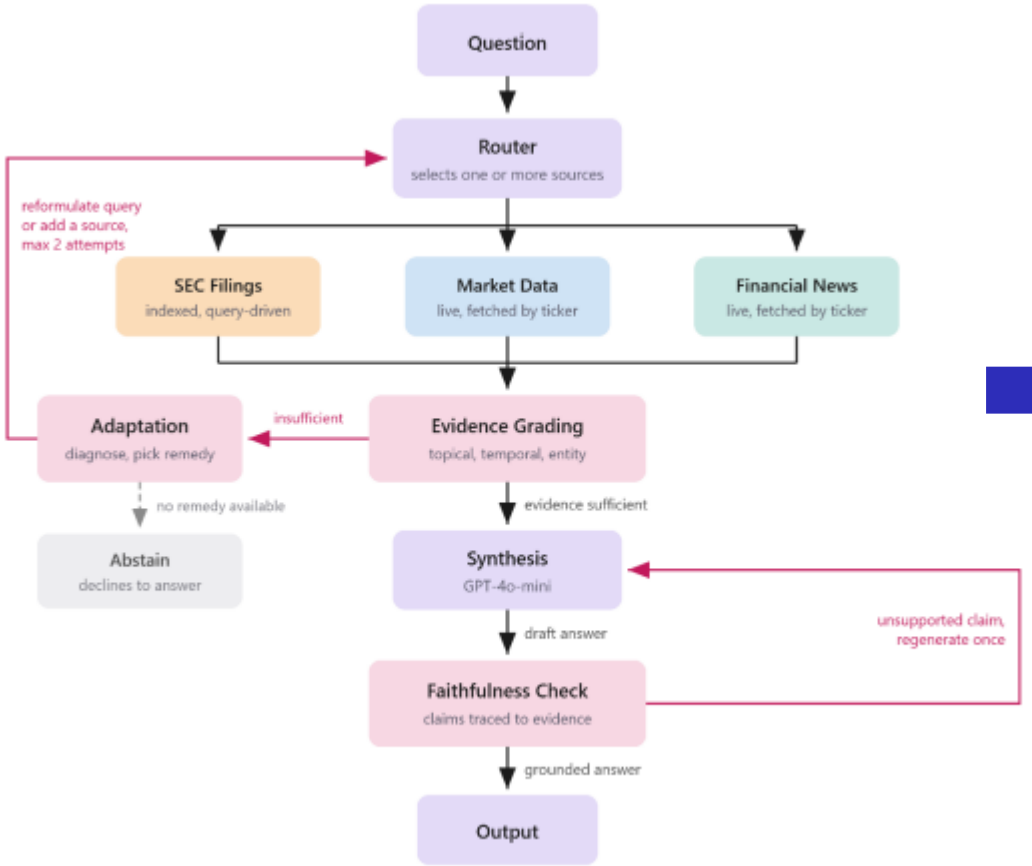
Career Interests and Plans
- Students at both the master's and doctoral degree levels were most interested in careers in industry. Doctoral degree–level students indicated they were most interested in academic careers at the start of their graduate studies.
- Graduate students at both the master's and doctoral degree levels feel that they have a limited knowledge of nonacademic careers.
- Career choices of graduate students are mostly influenced by their motivation to find work/life balance and their self-awareness. There is a notable difference between how various racial and ethnic groups tend to rate the importance of values in contributing to career decisions.
- The majority of graduate students feel the best time to learn about career areas and opportunities for graduate degree holders is before starting graduate school.
- The majority of graduate students rate search engines (e.g., Google) and professional conferences as the most useful career resources they use.

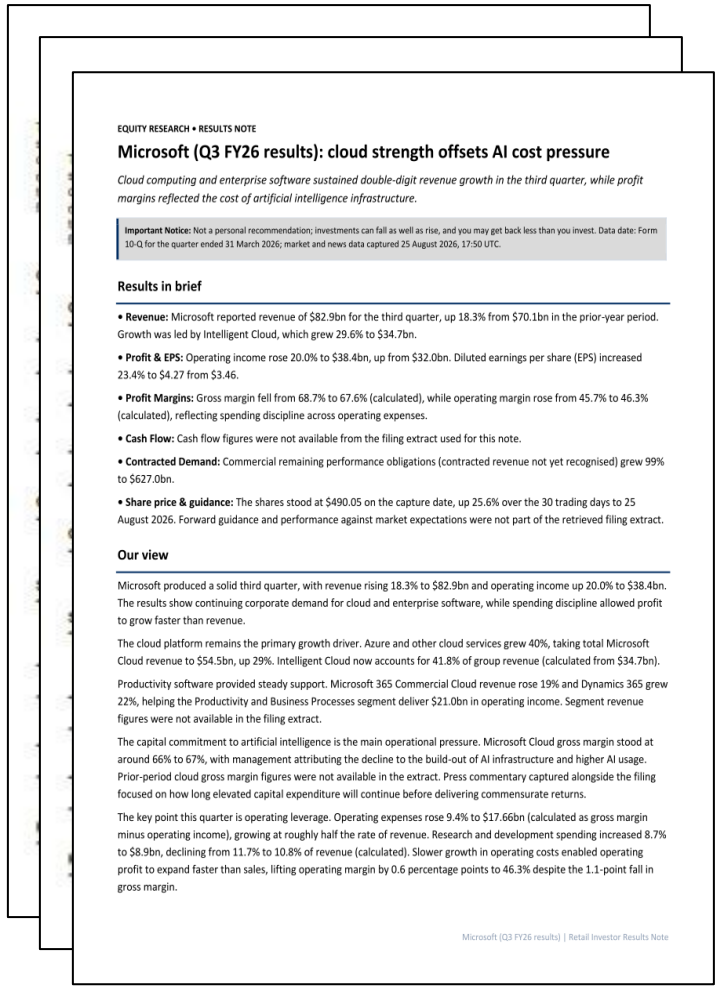
Graduate Education Competencies
- Majority of master's and doctoral degree students believe they are proficient in the core competency areas described by the National Academies Press report on *Graduate STEM Education for the 21st Century* (2018).[1]

Social/Contextual Factors
- Positive characteristic behaviors that students cited as descriptive of their research advisor center around providing an appropriate level of research independence and recognition (e.g., "encourages me to gain independence over the course of my graduate studies," and "gives the appropriate level of credit to me for my research contributions").
- Positive characteristic behaviors that students cited as less descriptive of their research advisor center around providing career path support and advice and taking a personal interest in them.
- Female research advisors, in comparison to male advisors, are perceived by graduate students as being more likely to encourage their students' engagement in professional development opportunities (e.g., writing grant proposals, presenting at conferences).
- Sixty-five percent of students reported that their advisor created an environment where group members were treated fairly.
- Having experienced harassment during their graduate school experience was indicated by 11% of female and 5% of male graduate students.

Resources and Benefits
- Many students are either not aware of or do not use available campus resources and benefits (e.g., safety training, career counseling, job placement services, etc.) available to them.

Dissertation: Agentic Financial Analysis





EQUITY RESEARCH • RESULTS NOTE

Microsoft (Q3 FY26 results): cloud strength offsets AI cost pressure

Cloud computing and enterprise software sustained double-digit revenue growth in the third quarter, while profit margins reflected the cost of artificial intelligence infrastructure.

Important Notice: Not a personal recommendation; investments can fall as well as rise, and you may get back less than you invest. Data date: Form 10-Q for the quarter ended 31 March 2026; market and news data captured 25 August 2026, 17:50 UTC.

Results in brief

- **Revenue:** Microsoft reported revenue of $82.9bn for the third quarter, up 18.3% from $70.1bn in the prior-year period. Growth was led by Intelligent Cloud, which grew 29.6% to $34.7bn.
- **Profit & EPS:** Operating income rose 20.0% to $38.4bn, up from $32.0bn. Diluted earnings per share (EPS) increased 23.4% to $4.27 from $3.46.
- **Profit Margins:** Gross margin fell from 68.7% to 67.6% (calculated), while operating margin rose from 45.7% to 46.3% (calculated), reflecting spending discipline across operating expenses.
- **Cash Flow:** Cash flow figures were not available from the filing extract used for this note.
- **Contracted Demand:** Commercial remaining performance obligations (contracted revenue not yet recognised) grew 99% to $627.0bn.
- **Share price & guidance:** The shares stood at $490.05 on the capture date, up 25.6% over the 30 trading days to 25 August 2026. Forward guidance and performance against market expectations were not part of the retrieved filing extract.

Our view

Microsoft produced a solid third quarter, with revenue rising 18.3% to $82.9bn and operating income up 20.0% to $38.4bn. The results show continuing corporate demand for cloud and enterprise software, while spending discipline allowed profit to grow faster than revenue.

The cloud platform remains the primary growth driver. Azure and other cloud services grew 40%, taking total Microsoft Cloud revenue to $54.5bn, up 29%. Intelligent Cloud now accounts for 41.8% of group revenue (calculated from $34.7bn).

Productivity software provided steady support. Microsoft 365 Commercial Cloud revenue rose 19% and Dynamics 365 grew 22%, helping the Productivity and Business Processes segment deliver $21.0bn in operating income. Segment revenue figures were not available in the filing extract.

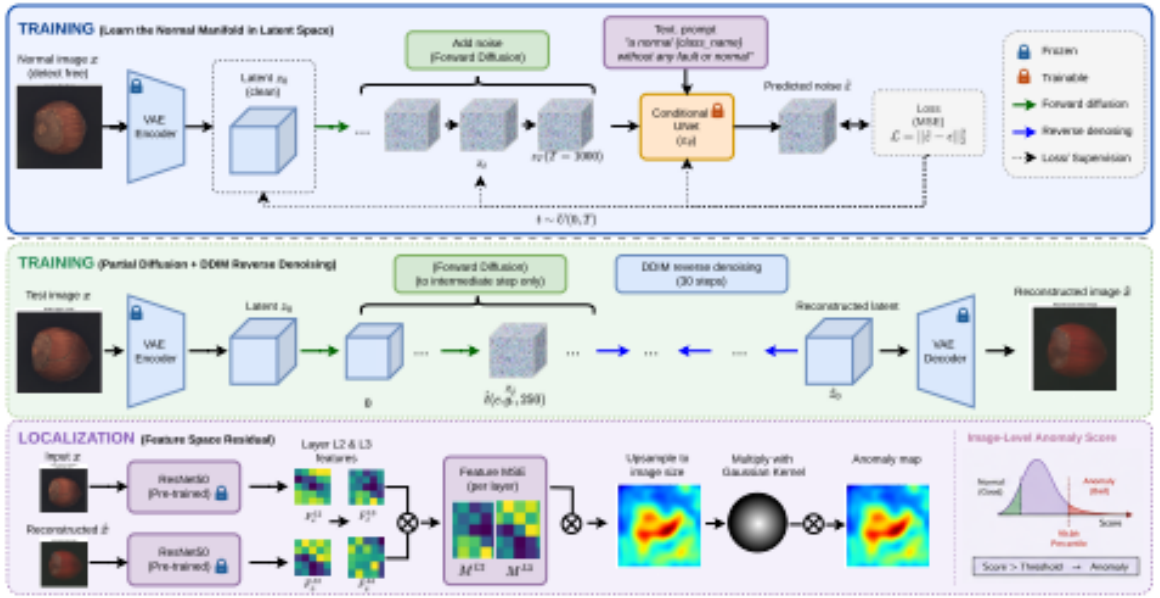
The capital commitment to artificial intelligence is the main operational pressure. Microsoft Cloud gross margin stood at around 66% to 67%, with management attributing the decline to the build-out of AI infrastructure and higher AI usage. Prior-period cloud gross margin figures were not available in the extract. Press commentary captured alongside the filing focused on how long elevated capital expenditure will continue before delivering commensurate returns.

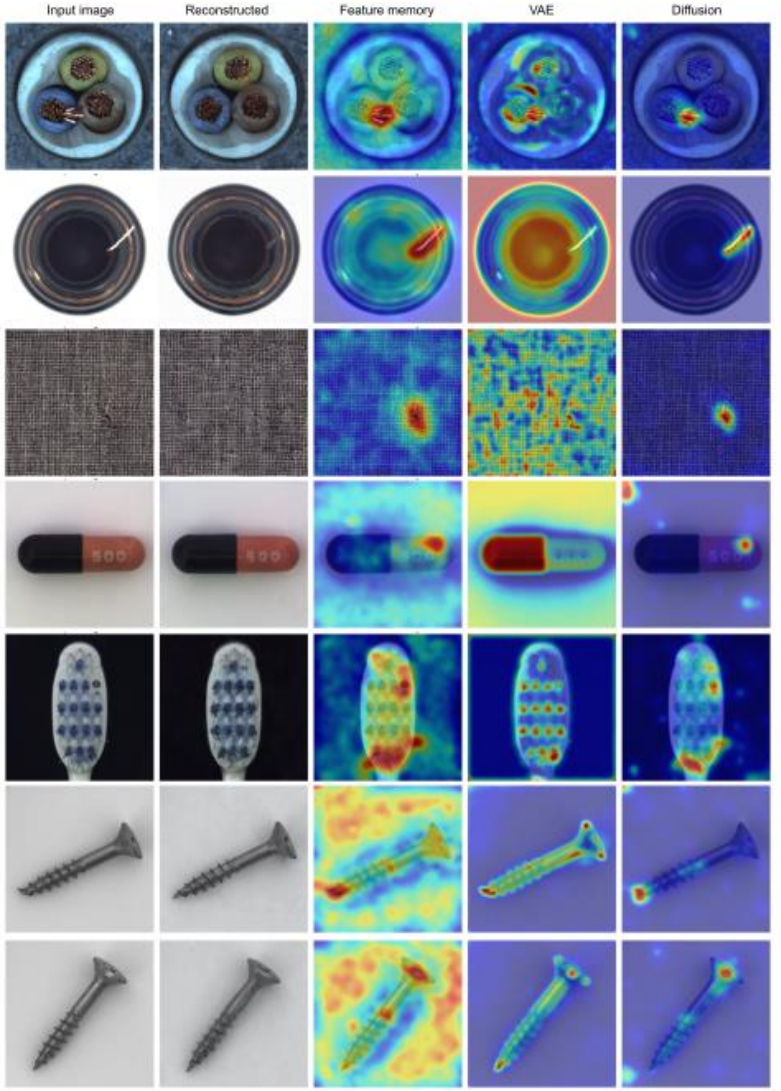
The key point this quarter is operating leverage. Operating expenses rose 9.4% to $17.66bn (calculated as gross margin minus operating income), growing at roughly half the rate of revenue. Research and development spending increased 8.7% to $8.9bn, declining from 11.7% to 10.8% of revenue (calculated). Slower growth in operating costs enabled operating profit to expand faster than sales, lifting operating margin by 0.6 percentage points to 46.3% despite the 1.1-point fall in gross margin.

Microsoft (Q3 FY26 results) | Retail Investor Results Note

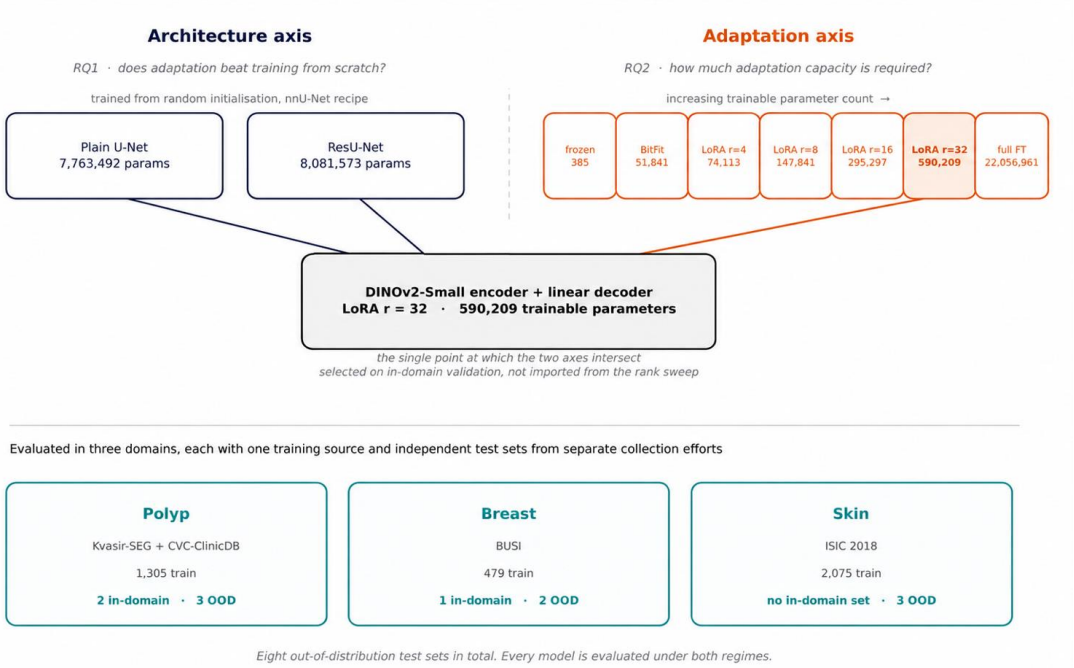
Dissertation: Industrial Anomaly Detection

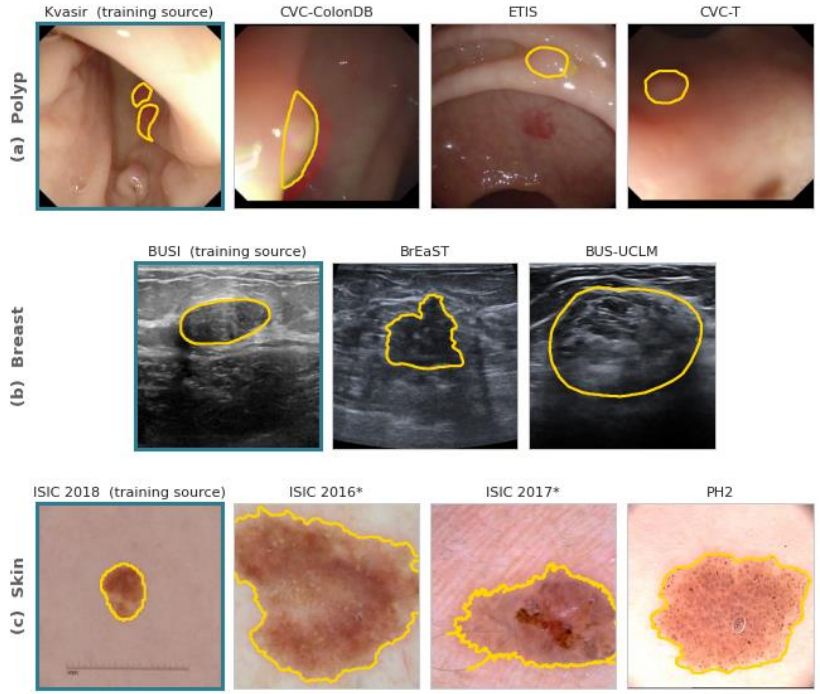
- Feature memory
- VAE models
- Diffusion models





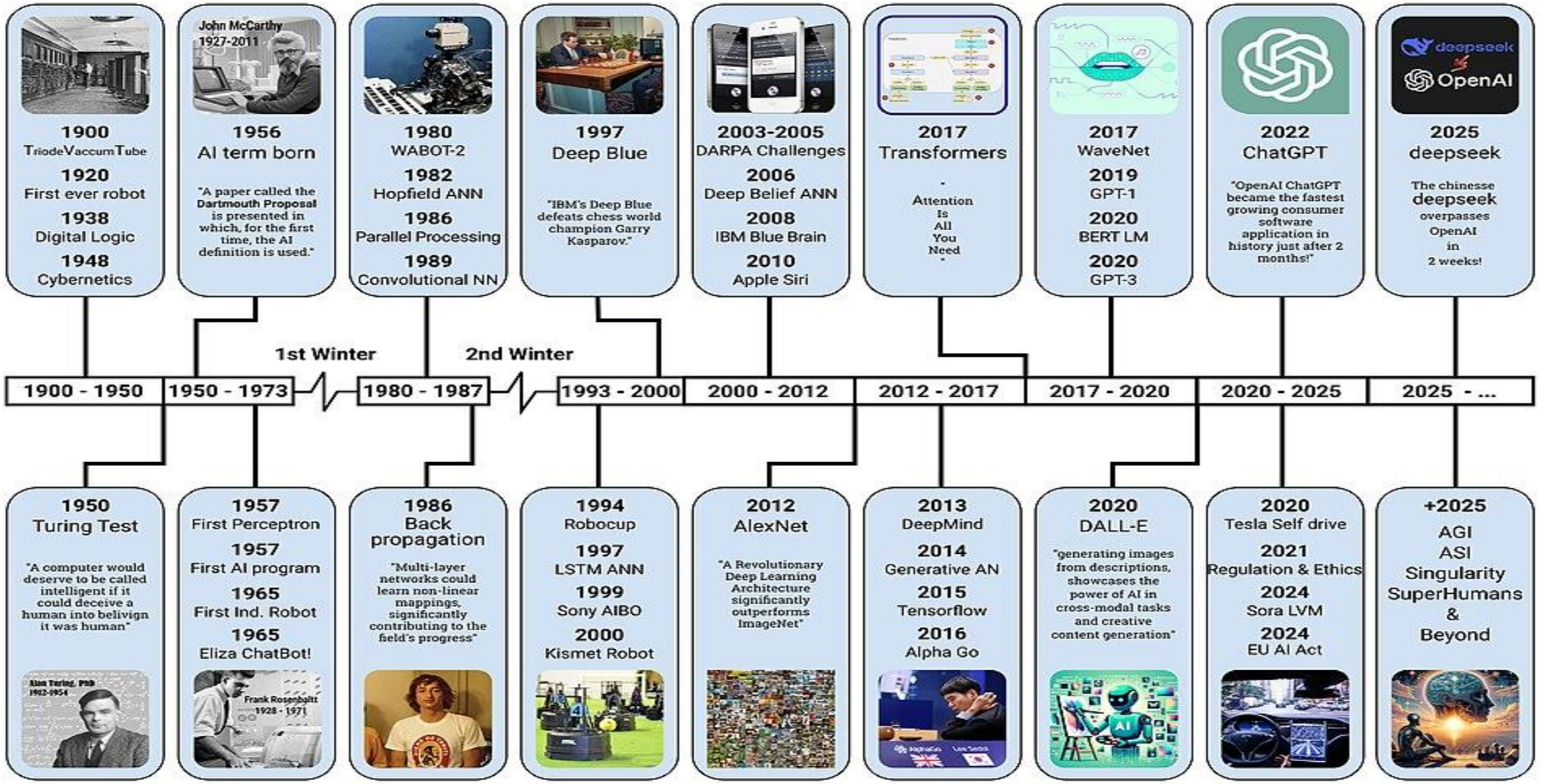
Dissertation: Model Adaptation of Medi-Images





- Specific models: UNet/ResUNet/nnUNet
- Adapted models:
  Frozen
  BitFit
  LoRA

History of AI



https://en.wikipedia.org/wiki/Timeline_of_artificial_intelligence

History of AI

Read…

Reading

[1] Russell, S. and Norvig, P. (2021). §1, Introduction.

[2] Negnevitsky M. (2011). §1, Introduction to knowledge-based intelligent system.